DUBLIN CITY UNIVERSITY

AUGUST/RESIT EXAMINATIONS 2016/2017

MODULE: CA208 - Logic

PROGRAMME(S):

CASE - BSc in Computer Applications (Sft.Eng.)
CPSSD - BSc in ComputationalProblem SolvandSW Dev.

YEAR OF STUDY: 2

EXAMINERS: David Sinclair (Ph:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	*Log Tables*	☐	*Thermodynamic Tables*
☐	*Graph Paper*	☐	*Actuarial Tables*
☐	*Dictionaries*	☐	*MCQ Only - Do not publish*
☐	*Statistical Tables*	☐	*Attached Answer Sheet*

Section A

QUESTION 1 ***[Total marks: 30]***

Consider the following problem.

- Either taxes are increased or if expenditures rise then the debt ceiling is raised.
- If taxes are increased, then the cost of collecting taxes rises.
- If a rise in expenditures implies that the government borrows more money, then if the debt ceiling is raised, then interest rates increase.
- If taxes are not increased and the cost of collecting taxes does not increase then if the debt ceiling is raised, then the government borrows more money.
- The cost of collecting taxes does not increase.
- Either interest rates do not increase or the government does not borrow more money.

1(a) [10 Marks]

Write the problem above as a set of propositional formulae.

1(b) [10 Marks]

Convert the propositional formulae from part (a) into Conjunctive Normal Form.

1(c) [10 Marks]

Using *Resolution*, prove that either the debt ceiling isn't raised or expenditures don't rise.

[End Question 1]

QUESTION 2 ***[Total marks: 10]***

[10 Marks]

Without using truth tables, show that

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q.$$

Clearly indicate which logical equivalence is used at each step.

[End Question 2]

QUESTION 3 ***[Total marks: 10]***

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$$

[End Question 3]

QUESTION 4 ***[Total marks: 10]***

[10 Marks]

Using Gentzen System $\mathcal{G}$ prove the following proposition is valid.

$$\neg(\neg p \vee q) \rightarrow (\neg q \wedge p)$$

[End Question 4]

QUESTION 5 ***[Total marks: 10]***

[10 Marks]

Using Gentzen System $\mathcal{G}$ prove the following first-order statement is valid.

$$\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$$

[End Question 5]

QUESTION 6 ***[Total marks: 10]***

[10 Marks]

Using *Skolemisation*, convert the following predicate expression into a set of clauses.

$$\forall x.((\exists y.R(x,y) \wedge \forall y.\neg S(x,y)) \rightarrow \neg(\exists y.R(x,y) \wedge P))$$

[End Question 6]

Section B

QUESTION 7 ***[Total marks: 10]***

When answering this question <u>do not use</u> any built-in list predicates.

7(a) [6 Marks]

Write a Prolog predicate `myReverse(X,Y)` that is true when the list `Y` is the reverse of list `X`.

7(b) [4 Marks]

Write a Prolog predicate `myDelete(X,Y,Z)` that is true when the list `Z` is the result of deleting an occurrence of the element `X` from the list `Y`, e.g. `myDelete(3, [1,2,3,4,3,2,1], [1,2,4,3,2,1])` is true.

[End Question 7]

QUESTION 8 ***[Total marks: 10]***

[10 Marks]

Write a Prolog predicate `oddsquaresum(X, S)` that is true when `S` is the sum of the square of all the odd numbers in the list `X`. Note that the list `X` can contain any atom, e.g. `[4,oranges,5,2,7,apples,1]`. Hence, `oddsquaresum([4,oranges,5,2,7,apples,1], 75)` is true.

[End Question 8]

QUESTION 9 ***[Total marks: 10]***

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

[End Question 9]

[APPENDICES]

Semantic Tableaux

α	α_1	α_2
$\neg\neg A_1$	A_1	
$A_1 \wedge A_2$	A_1	A_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$

β	β_1	β_2
$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$B_1 \vee B_2$	B_1	B_2
$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \downarrow B_2)$	B_1	B_2
$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$

γ	$\gamma(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U_1' \cup \{\alpha_1, \alpha_2\}}{\vdash U_1' \cup \{\alpha\}} \qquad \frac{\vdash U_1' \cup \{\beta_1\} \qquad \vdash U_2' \cup \{\beta_2\}}{\vdash U_1' \cup U_2' \cup \{\beta\}}$$

$$\frac{\vdash U_1' \cup \{\gamma, \gamma(a)\}}{\vdash U_1' \cup \{\gamma\}} \qquad \frac{\vdash U_1' \cup \{\delta(a)\}}{\vdash U_1' \cup \{\delta\}}$$

$\dfrac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2
$\neg\neg A$	A	
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$
$A_1 \vee A_2$	A_1	A_2
$A_1 \rightarrow A_2$	$\neg A_1$	A_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$

β	β_1	β_2
$B_1 \wedge B_2$	B_1	B_2
$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$\neg(B_1 \uparrow B_2)$	B_1	B_2
$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$

γ	$\gamma(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]